

Michael Chen

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portfolio-five-ecru-5woj03hk11.vercel.app

EDUCATION

University of Washington

Master of Science in AI/ML for Engineering

Expected 2028

University of Illinois Urbana-Champaign

Bachelor of Science in Physics – Computational Physics Professional Track

May 2026

- GPA: 3.75
- Relevant Coursework: Machine Learning, Quantum Computing, Data Structures, Numerical Methods

EXPERIENCE

IDX Exchange

Data Analyst Intern

Jun. 2026 – Aug. 2026

- Cleaned and unified monthly MLS listing and sold datasets sourced through the CoreLogic Trestle API, applying date-consistency checks, geographic validation, and tiered IQR outlier flagging across 200K+ records.
- Enriched transaction data with FRED 30-year mortgage rates, resampling weekly observations to monthly values and merging them on a year-month key with null and consistency validation.
- Engineered analytical features including price ratio, price per square foot, listing-to-contract days, and contract-to-close days, and added California school-district boundaries through GeoPandas spatial joins.
- Built two Tableau workbooks for market and competitive analysis, including top-100 agent and brokerage rankings, ZIP-code heat maps, and interactive filters; presented findings through a market intelligence report and live presentation.

MACHINE LEARNING & DATA SCIENCE PROJECTS

Galaxy Zoo Galaxy Classification | *Python, PyTorch, CNNs, NumPy*

- Developed a convolutional neural network in PyTorch to classify galaxy morphology from astronomical image data using the Galaxy Zoo dataset.
- Built an end-to-end image classification pipeline with preprocessing, convolutional layers, pooling, optimization, and model evaluation.

Hopfield Networks & Restricted Boltzmann Machines | *Python, NumPy*

- Implemented Hopfield Networks and Restricted Boltzmann Machines from scratch using Python and NumPy without high-level ML frameworks.
- Developed network-state updates, energy-based optimization, probabilistic sampling, and weight-learning algorithms for associative and generative tasks.

Quantum Computing Simulator & Shor's Algorithm | *Python, NumPy*

- Built a quantum-computing simulator modeling quantum states, gates, measurements, and multi-qubit operations using linear algebra.
- Implemented components of Shor's algorithm to simulate quantum period finding and integer factorization.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Generative Models, Energy-Based Models

Data Science: pandas, NumPy, Matplotlib, GeoPandas, Feature Engineering, EDA, Statistical Analysis

Tools: Tableau, Git, GitHub, Jupyter, VS Code, Excel